Camouflage Unlearning: Fast and Surgical Machine Unlearning for Biometric Systems

***Abstract*—Machine unlearning for privacy-critical biometric systems such as fingerprint, iris, and face recognition remains underexplored, despite the growing need for efficient solutions under General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) mandates that require data erasure requests to be supported throughout a model's deployment lifecycle. Existing methods predominantly target classification and generative tasks, and rely on many-to-one feature collapse that forces all forget classes toward a single distant point in feature space. This increases convergence time and computational cost, making them impractical for resource-constrained edge deployments where biometric models face frequent sequential removal requests with minimal replay data. Additionally, existing methods have been benchmarked almost exclusively on ResNet or ViT backbones for CIFAR and ImageNet, leaving their effectiveness on biometric backbones largely untested. In this work, we propose Camouflage Unlearning (CU), a cycle-robust framework that maps forget classes toward nearby retain-class boundaries rather than collapsing to a single point, thereby reducing migration distance. CU achieves 3–4× faster convergence through retain-dominance constraints and localized regularization that performs surgical updates in specific transformer blocks, while maintaining competitive retain accuracy. Evaluation across four domains, classification, iris, fingerprint, and face recognition demonstrates competitive performance and privacy behavior (MIA ≈ 0.5), supporting CU's effectiveness for biometric authentication systems requiring regulatory compliance.**

***Index Terms*—Biometric recognition, data privacy, face recognition, GDPR, machine unlearning, vision transformers**

I. Introduction

MACHINE unlearning (MU) has emerged as a critical research area, addressing the imperative to prevent unauthorized data exploitation, eliminate harmful model behaviors, and safeguard user privacy [1]. A natural consideration follows: what drives the necessity for unlearning? The proliferation of AI misuse ranging from phishing generation and model inversion attacks to perpetuating misinformation has prompted regulatory frameworks [2], [3], [4]. Frameworks like the General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) [5], [6] mandate data erasure capabilities. These regulations require model service providers to erase user data from learned representations upon request. This establishes both legal and ethical obligations for responsible AI deployment. Crucially, users possess the right to request data removal at any time. This transforms unlearning from a one time operation into a continuous process throughout a model's deployment lifecycle. Per-request efficiency, therefore, becomes a first-class concern.

Existing machine unlearning methods predominantly target image classification [7] and generative tasks [8], with limited exploration of privacy-critical biometric systems. While face recognition [9], [10], [11] has received modest attention, fingerprint and iris recognition remains substantially less studied. This gap persists despite their sensitive nature and widespread deployment in authentication systems. Contemporary approaches rely primarily on KL divergence maximization [12] or gradient ascent [13]. These methods implement many-to-one mappings that force forgotten class embeddings toward a single distant point in feature space. Alternative strategies include saliency-based selection [14], geometric boundary manipulation [15], perturbation-based forgetting [16], and meta-learning optimization [17], [18]. A key observation across all these methods is that every forget sample is driven toward a single collapse point in the decision space. Even when a sample already lies far from the retain manifold, it must still traverse the full distance to this distant target. This increases convergence time and GPU cost, as observed in Table I. This many-to-one collapse behavior is visualized and ablated in detail in Section III-B. Concretely, GS-LoRA [13] (group-sparse LoRA, slow ascent), SCRUB [12] (bidirectional logits, unstable KL), SalUn [14] (weight saliency, reversible labels), Boundary Unlearning [15] (shadow-node pruning, slow), Forget Vectors [16] (noise tuning, invertible), MCU [18] (Bézier curves, dual optimization), LTU [17] (meta-learning, computationally expensive), DELETE [19] and LoTUS [20] (frozen teacher, double memory), and MUNBa [21] (Nash bargaining, per-step overhead) each trade convergence speed, memory, or reversibility, and few explicitly target biometric edge deployment or sequential forget cycles.

Biometric recognition differs from general classification in ways that directly affect unlearning feasibility. Biometric systems deploy on resource-constrained edge devices. Removal requests arrive sequentially and frequently over the deployment lifecycle. This transforms unlearning into a recurring operational cost rather than a one-time procedure. Under this workload, existing many-to-one methods become prohibitive as per request compute scales poorly with request frequency. Compounding this, existing unlearning methods have been designed and benchmarked almost exclusively on ResNet [22] or ViT backbones [23] for CIFAR [24] and ImageNet [25]. Their transferability to biometric backbones such as Face Transformer [26] (sliding-patch ViT face recognition), LAM [27] (ResNet-50, lightweight attention), JIPNet [28] (CNN-Transformer joint pose-identity), SSLDMNet-Iris [29] (CNN-GRU, Fisher discriminant), IrisFormer [30] (rotation-invariant ViT matching), PFSimNet [31] (partial fingerprint similarity network), and MEM-FR [32] (optical, privacy-preserving recognition) remains untested, despite these architectures being the standard in deployed authentication systems. What biometric unlearning requires instead is a cycle-robust method validated directly on biometric backbones.

To address these challenges, we propose Camouflage Unlearning (CU), a novel framework that achieves 3–4× faster convergence while maintaining robust performance. Unlike existing many-to-one approaches, CU pushes forget-class embeddings toward the nearest retained class boundaries, effectively reducing migration distances. To perform surgical modifications and minimize collateral damage to retain classes, CU employs localized regularization concentrated in specific blocks of the neural network for additional details, please refer to Section II.

Our main contributions are summarized as follows:

1) To the best of our knowledge, this is the among the earliest comprehensive studies of machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition.
2) We introduce Camouflage Unlearning, which achieves 3–4× faster convergence compared to SoTA unlearning methods while maintaining competitive retention accuracy and privacy guarantees.
3) We conduct extensive empirical evaluation across multiple biometric and classification domains, demonstrating that CU concentrates updates in specific network blocks and remains effective under data-scarce replay conditions.

II. Problem Setting and Methodology

A. Problem Setting

Let $\mathcal{M}$ be a model pre-trained on $D = D_r^{full} \cup D_f$ ($D_r^{full} \cap D_f = \emptyset$), representing a mapping $f_\mathcal{M} : \mathcal{X}_D \to \mathcal{Y}_D$. Storing D_r^{full} is prohibited under GDPR/CCPA, and full retraining is costly, so we retain only a small replay buffer $D_r \subset D_r^{full}$, with $|D_r|$ limited to at most 10% of $|D_r^{full}|$[1]. Before unlearning, $\mathcal{M}$ correctly maps both D_f and D_r. Applying an unlearning algorithm $\mathcal{F}$ (Section II-B) yields $\mathcal{M}' = \mathcal{F}(\mathcal{M}, D_f, D_r)$, under which the forget mapping should no longer hold while the retain mapping persists:

$$f_\mathcal{M} : \mathcal{X}_{D_f} \to \mathcal{Y}_{D_f} \,;\, \mathcal{X}_{D_r} \to \mathcal{Y}_{D_r} \implies f_{\mathcal{M}'} : \mathcal{X}_{D_f} \not\to \mathcal{Y}_{D_f} \,;\, \mathcal{X}_{D_r} \to \mathcal{Y}_{D_r} \tag{1}$$

where $\not\to$ denotes a forgotten mapping and $\to$ a retained one. We define successful unlearning as

- Forgetting: $Acc(\mathcal{M}', D_f) \leq \frac{1}{C}$,
- Retention: $Acc(\mathcal{M}', D_r) \approx Acc(Oracle, D_r)$

where C is the number of classes and Oracle denotes a model trained directly on target classes only.

B. Camouflage Unlearning

CU relocates forget-class embeddings to retain-class boundaries via direct centroid mapping. Unlike existing many-to-one approaches that collapse all forget classes to a single distant point, CU implements many-to-many distribution, reducing migration distance and accelerating convergence.

We compute class centroids by averaging embeddings, $c_k = \frac{1}{|D_k|} \sum_{x \in D_k} emb(x)$, yielding retain centroids $\{c_{r_1}, \ldots, c_{r_K}\}$ and forget centroids $\{c_{f_1}, \ldots, c_{f_M}\}$, where K and M are the numbers of retain and forget classes. Each forget centroid c_{f_i} is assigned its nearest retain centroid as target:

$$\mathbf{t}_i = c_{r_{j^*}}, \quad j^* = \arg\min_j \|c_{f_i} - c_{r_j}\|_2 \tag{2}$$

directly mapping forget embeddings into retain decision boundaries and minimizing migration distance. We pull forget embeddings toward their assigned targets:

$$\mathcal{L}_f = \frac{1}{|D_f|} \sum_{(\mathbf{x},y) \in D_f} \|emb(\mathbf{x}) - \mathbf{t}_y\|^2 \tag{3}$$

which collapses forget classes into retain boundaries but risks corrupting retain decision regions and prolonging convergence. We therefore enforce that retain logits dominate forget logits by margin $\gamma = 2.0$:

$$\mathcal{L}_{dom} = ReLU\left(\max_{i \in F} z_i + \gamma - \max_{j \in R} z_j\right) \tag{4}$$

where z_i are logits and F, R denote forget and retain class indices; this accelerates forget-class suppression while preserving retain accuracy. Retain-class knowledge is preserved through classification and embedding stability:

$$\mathcal{L}_{ret} = \lambda_{ce} \cdot CE(\mathbf{z}_r, y_r) + \lambda_{emb} \cdot \frac{1}{|D_r|} \sum_{(\mathbf{x},y) \in D_r} \|emb(\mathbf{x}) - c_y\|^2 \tag{5}$$

where $\mathbf{z}_r$ are retain logits, y_r are labels, and c_y are original retain centroids. Since comprehensive parameter updates risk instability under limited replay data, inspired by group sparse selection [13], we apply localized regularization that selectively updates fewer transformer blocks:

$$\mathcal{L}_{local} = \lambda_{local} \sum_{g=1}^{G} \|\theta_g\|_2 \tag{6}$$

where θ_g represents parameters in block g and G is the total number of groups. Each θ_g is initialized at zero and tracks the *change* accumulated in block g relative to the pretrained weights, so $\mathcal{L}_{local}$ penalizes the magnitude of updates rather than absolute parameter magnitude. This group-Lasso-inspired approach concentrates updates in specific blocks while leaving others unchanged, preserving retained knowledge under constrained replay. The total loss integrates all components:

$$\mathcal{L}_{total} = \mathcal{L}_f + \lambda_{dom}\mathcal{L}_{dom} + \mathcal{L}_{ret} + \mathcal{L}_{local} \tag{7}$$

III. Performance Evaluation

A. Experimental Setup

Datasets, Backbones, and Baselines: We evaluate on CIFAR-100 [24] (DeiT-Tiny [33]), CASIA-Iris [34] (LAM [27]), Sokoto Coventry fingerprints [35] (JIPNet [28]), and CASIA-Face100 [36] (Face Transformer [26]), comparing CU against SCRUB [12], SalUn [14], GS-LoRA [13], LTU [17], MCU [18], and Forget Vectors [16], plus an Oracle trained only on $D_r^{\mathbf{full}}$ as the upper bound.

Evaluation Metrics: We report Acc_f, Acc_r, MIA (label-only attack [37], ≈ 0.5 = safe), KL divergence to Oracle, RTE, and H-Mean [38]. Success requires $Acc_f \approx 0$, $Acc_r \approx$ Oracle,

[1] From this point, D_r denotes the replay buffer unless specified.

Method	Classification						Iris recognition					
	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓
Oracle	–	77.36	–	–	–	–	–	99.03	–	–	–	–
GS-LoRA [13]	0.11	71.32	74.18	0.73	0.56	153.2	0.00	93.42	96.18	0.75	0.69	225.7
SalUn [14]	0.37	72.46	74.63	1.32	0.63	225.6	0.37	92.37	95.44	0.79	0.59	427.2
SCRUB [12]	0.58	73.11	74.90	1.89	0.64	219.3	0.43	95.58	97.07	0.83	0.63	501.8
FV [16]	0.00	74.43	75.89	0.87	0.55	189.5	0.00	96.27	97.63	1.13	0.57	383.1
MCU [18]	0.42	74.43	75.66	0.87	0.59	175.5	0.00	98.87	98.95	0.99	0.63	437.2
LTU [17]	0.00	76.32	76.83	0.75	0.57	239.3	0.73	95.43	96.85	1.25	0.55	329.4
CU	0.47	74.46	75.65	1.27	0.56	55.5	0.06	97.89	98.43	1.01	0.53	116.1

Method	Fingerprint recognition						Face recognition					
	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓
Oracle	–	90.82	–	–	–	–	–	95.12	–	–	–	–
GS-LoRA [13]	0.00	83.42	86.98	1.13	0.65	155.3	0.57	93.25	93.90	1.13	0.53	284.4
SalUn [14]	0.10	82.32	86.24	0.75	0.63	202.2	0.00	90.15	92.56	1.27	0.59	417.2
SCRUB [12]	0.00	83.27	86.85	2.36	0.57	175.3	0.42	93.27	93.98	2.87	0.63	389.5
FV [16]	0.30	84.44	87.38	1.27	0.54	160.7	0.20	92.27	93.58	1.20	0.60	380.3
MCU [18]	0.00	80.32	85.24	0.99	0.59	220.5	0.00	90.12	92.54	0.89	0.57	403.8
LTU [17]	0.00	87.72	89.24	1.13	0.54	237.4	0.00	92.58	93.84	0.93	0.55	430.2
CU	0.00	87.23	88.98	1.05	0.55	55.32	0.00	93.98	94.55	1.67	0.51	175.3

TABLE I
PERFORMANCE ACROSS FOUR DOMAINS; CU ACHIEVES 3–4× RTE SPEEDUP.

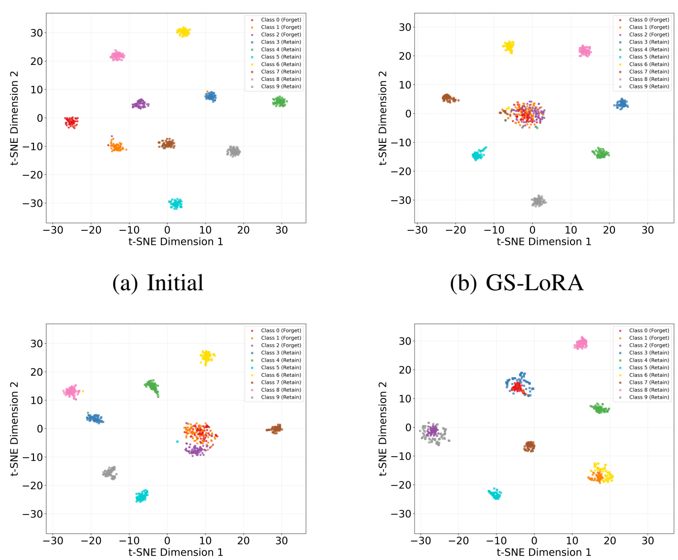
low KL/RTE. FAR@$\{0.3, 0.4, 0.5\}$/EER (cosine similarity, forget probes vs. retain gallery) further verify open-set identity confusion (Section III-D).

B. Main Results

Table I compares all methods across four tasks; green/blue mark best/second-best.

Results: All methods forget near-perfectly ($Acc_f < 1\%$). CU ranks first or second in Acc_r and H-Mean across all tasks (LTU and MCU the closest competitors), and achieves MIA ≈ 0.5 despite GS-LoRA's lower KL, expected since CU deliberately reshapes forget logits. Most critically, CU delivers a 3–4× RTE speedup over every baseline.

Geometric Analysis: Figure 1 visualizes embedding-space t-SNE. GS-LoRA [13], SCRUB [12], and Boundary Unlearning [15] collapse forget classes to one distant point; CU instead migrates them to the nearest retain boundary, explaining its RTE gains in Table I.



(a) Initial (b) GS-LoRA

(c) Boundary (d) CU (Ours)

Fig. 1. t-SNE: CU spreads embeddings; baselines collapse to one point.

C. Ablation Studies

Replay Buffer Size: We investigate the minimal replay buffer size $|D_\mathbf{r}|$ for effective unlearning on CIFAR-100. Table II shows that a 0.1 ratio optimally balances retention accuracy and efficiency.

Buffer Ratio	Speed	Acc_f	Acc_r	H-$Mean$
1.0	1×	1.33	74.39	2.62
0.5	2×	1.18	73.00	2.32
0.3	3.3×	1.33	74.55	2.61
0.1	10×	1.30	74.52	2.56

TABLE II
REPLAY BUFFER SIZE ABLATION; 0.1 RATIO BALANCES RETENTION/EFFICIENCY.

Retain Dominance Constraint: Without this constraint, we found that the forget accuracy exhibits a prolonged tail, delaying convergence. Table III demonstrates that applying the constraint reduces Acc_f from 9.97% to 0.39%, achieving faster unlearning with minimal retention loss.

Method	Acc_f	Acc_r
Without Constraint	9.97	78.12
With Constraint	0.39	77.06

TABLE III
DOMINANCE CONSTRAINT ELIMINATES LONG-TAIL CONVERGENCE.

Localized Parameter Regularization: This regularization selectively modifies fewer blocks, enabling surgical edits that preserve model capacity. On CIFAR-100 classification, Figure 2 shows concentrated updates in specific blocks, while Table IV demonstrates reducing total ℓ_2 norm from 39.05 to 10.09 with minimal retention loss.

Method	Acc_f	Acc_r	Total ℓ_2 Norm
Without Localized Reg	0.00	77.24	39.05
With Localized Reg	0.00	74.76	10.09

TABLE IV
LOCALIZED REGULARIZATION LOWERS ℓ_2 NORM; MINIMAL ACCURACY LOSS.

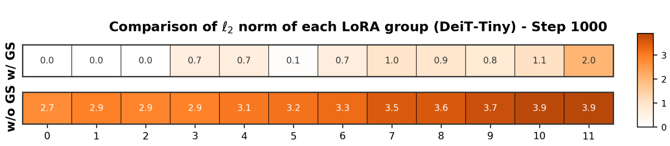


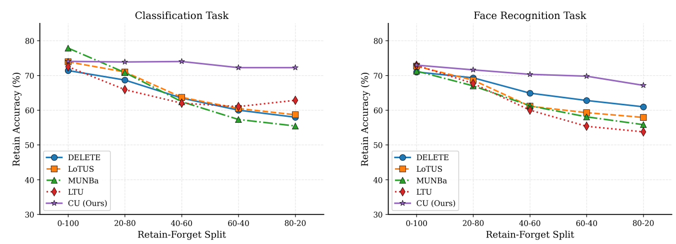
Fig. 2. Per-block ℓ_2 norm: updates concentrate in few blocks.

Scalability: We apply CU to five ViT backbones (DeiT-Tiny to ViT-Large). Table V shows consistent forgetting and retention across both lightweight and large-scale architectures.

Model	Before Unlearning		After Unlearning	
	Acc_f	Acc_r	$Acc_f \downarrow$	$Acc_r \uparrow$
DeiT-Tiny	75.25	75.67	0.40	74.46
DeiT-Small	74.86	74.01	1.35	73.89
DeiT-Base	74.32	74.15	0.87	72.58
ViT-Base	72.58	73.21	2.35	70.35
ViT-Large	75.35	78.91	0.10	75.61

TABLE V
VIT BACKBONE SCALE VS. UNLEARNING PERFORMANCE.

Sequential Removal Workload: We pilot CU against DELETE [19], LoTUS [20], MUNBa [21], and LTU [17] across successive forget cycles on CIFAR-100 [24] and CASIA-Face100 [36]. Figure 3 shows baselines degrade progressively while CU retains stable accuracy, confirming robustness under continual removal.



(a) CIFAR-100 (b) Face Recognition

Fig. 3. Sequential-cycle retain accuracy; baselines degrade, CU stable.

D. Attack-Based Verification

Table VI reports FAR/EER of forget-class probes matched against the retain gallery on CASIA-Iris. CU tracks the Oracle closely (FAR@0.5 3.83% vs. 3.10%; EER 24.99% vs. 21.69%), and both rise similarly over *Before* unlearning (1.64%), indicating that the small increase in open-set false accepts stems from removing the identities rather than from CU specifically.

Model	FAR@0.3	FAR@0.4	FAR@0.5	EER
Before	11.38%	4.84%	1.64%	13.33%
CU	17.97%	8.52%	3.83%	21.99%
Oracle	16.10%	7.79%	3.10%	21.69%

TABLE VI
FAR/EER OF FORGET PROBES VS. RETAIN GALLERY (CASIA-IRIS).

Attention Maps: We visualize attention on forget/retain CASIA-Face100 [36] samples before/after CU. Figure 4 shows the base model attends to facial regions for both; retrained and CU models shift attention to peripheral areas *only for forget samples*, confirming CU surgically removes identity-specific features while preserving retain ones.

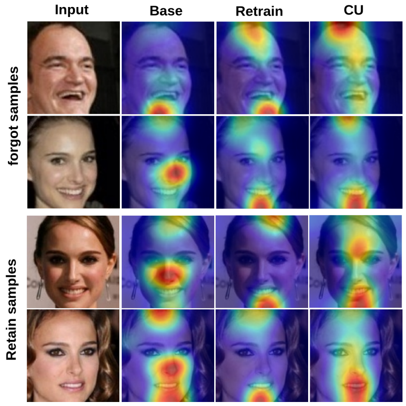


Fig. 4. Attention maps: retrained/CU models disperse attention off-face only for forget identities.

Relearning Attacks: We test whether CU's forgetting is genuine or reversible via a relearning attack: fine-tuning a CU-unlearned model (50 classes) and a head-masked baseline on forgotten classes, tracking recovery. Figure 5 shows CU recovers gradually versus head-masking's rapid recovery, confirming genuine backbone-level forgetting.

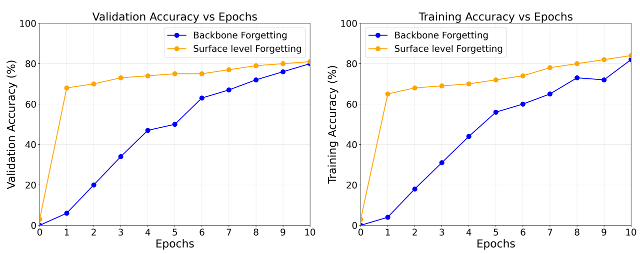


Fig. 5. CU recovers gradually (genuine) vs. rapid head-masking (superficial).

IV. Results Discussion

A. On the Necessity of the Retain Buffer

A natural concern is whether the replay buffer $D_r \subset D_r^{\text{full}}$ contradicts GDPR/CCPA erasure mandates. We argue D_r is unavoidable: pretraining jointly entangles D_f and D_r^{full} in the same parameters, so disentangling forget/retain knowledge without any retained reference is infeasible without architectural support such as modular pathways. Existing alternatives fall short: impair-and-repair [39] still needs D_r for repair, source-free unlearning [40] works only for convex/linear models, and all established non-convex methods, via Fisher/saliency [41], [14], retention losses [12], [13], or continual mechanisms [38], [42], [43], [44], [45], rely on D_r; only generative-task exceptions like FLAT [46] and ESD [47] avoid it. CU therefore keeps a minimal D_r, at most 10% of D_r^{full}, at least one sample per retain identity, consistent with limited lawful processing under GDPR/CCPA. Retain metrics are computed on the full held-out test split, not on D_r itself. Crucially, D_r holds only *consenting* retain identities: a withdrawal is treated as a forget request, and the buffer is refilled from the remaining consenting pool.

V. Conclusion

To our knowledge, we are among the earliest to address machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition. We observed that existing methods suffer from prolonged convergence due to many-to-one embedding collapse, which is particularly inefficient for biometric models experiencing frequent unlearning requests on edge devices. To address this, we propose Camouflage Unlearning, achieving up to 4× speedup through many-to-many distribution and surgical parameter modifications while delivering competitive forgetting, retention, and privacy performance comparable to state-of-the-art methods. Future work includes extending Camouflage Unlearning to continual forgetting scenarios in pretrained biometric models.

References

[1] D. Zhu, D. Chen, X. Wu, J. Geng, Z. Li, J. Grossklags, and L. Ma, "Privauditor: Benchmarking data protection vulnerabilities in llm adaptation techniques," *Advances in Neural Information Processing Systems*, vol. 37, pp. 9668–9689, 2024.
[2] O. Li, Y. Hao, Z. Wang, B. Zhu, S. Wang, Z. Zhang, and F. Feng, "Model inversion attacks through target-specific conditional diffusion models," *arXiv preprint arXiv:2407.11424*, 2024.
[3] M. Prabhakar, L. Xu, and P. Saxena, "Model inversion attacks meet cryptographic fuzzy extractors," *arXiv preprint arXiv:2510.25687*, 2025.
[4] M. Kahla, S. Chen, H. A. Just, and R. Jia, "Label-only model inversion attacks via boundary repulsion," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 15 045–15 053.
[5] E. Goldman, "An introduction to the california consumer privacy act (ccpa)," *Santa Clara Univ. Legal Studies Research Paper*, 2020.
[6] G. Data, "General data protection regulation (gdpr)," *Intersoft Consulting, Accessed in October*, vol. 24, no. 1, 2018.
[7] A. Wuerkaixi, S. Cui, J. Zhang, K. Yan, B. Han, G. Niu, L. Fang, C. Zhang, and M. Sugiyama, "Accurate forgetting for heterogeneous federated continual learning," *arXiv preprint arXiv:2502.14205*, 2025.
[8] L. Wei, Y. Nakashima, and N. Garcia, "Emma: Concept erasure benchmark with comprehensive semantic metrics and diverse categories," *arXiv preprint arXiv:2512.17320*, 2025.
[9] T. De Min, M. Mancini, S. Lathuilière, S. Roy, and E. Ricci, "Unlearning personal data from a single image," *arXiv preprint arXiv:2407.12069*, 2024.
[10] M. Zakharov, "Face identity unlearning for retrieval via embedding dispersion," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2026, pp. 1277–1286.
[11] F. Shivam, N. Najafzadeh, Y. Reddy, and P. Gyawali, "Cure: Centroid-guided unsupervised representation erasure for facial recognition systems," *arXiv preprint arXiv:2509.19562*, 2025.
[12] M. Kurmanji, P. Triantafillou, J. Hayes, and E. Triantafillou, "Towards unbounded machine unlearning," *Advances in neural information processing systems*, vol. 36, pp. 1957–1987, 2023.
[13] H. Zhao, B. Ni, J. Fan, Y. Wang, Y. Chen, G. Meng, and Z. Zhang, "Continual forgetting for pre-trained vision models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 28 631–28 642.
[14] C. Fan, J. Liu, Y. Zhang, E. Wong, D. Wei, and S. Liu, "Salun: Empowering machine unlearning via gradient-based weight saliency in both image classification and generation," *arXiv preprint arXiv:2310.12508*, 2023.
[15] M. Chen, W. Gao, G. Liu, K. Peng, and C. Wang, "Boundary unlearning," *arXiv preprint arXiv:2303.11570*, 2023.
[16] C. Sun, R. Wang, Y. Zhang, J. Jia, J. Liu, G. Liu, Y. Yan, and S. Liu, "Forget vectors at play: Universal input perturbations driving machine unlearning in image classification," *arXiv preprint arXiv:2412.16780*, 2024.
[17] M. H. Huang, L. G. Foo, and J. Liu, "Learning to unlearn for robust machine unlearning," in *European Conference on Computer Vision*. Springer, 2024, pp. 202–219.
[18] Y. Shi and R. Wang, "Mcu: Improving machine unlearning through mode connectivity," *arXiv preprint arXiv:2505.10859*, 2025.

[19] Y. Zhou, D. Zheng, Q. Mo, R. Lu, K.-Y. Lin, and W.-S. Zheng, "Decoupled distillation to erase: A general unlearning method for any class-centric tasks," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 20 350–20 359.

[20] C. N. Spartalis, T. Semertzidis, E. Gavves, and P. Daras, "Lotus: Large-scale machine unlearning with a taste of uncertainty," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 10 046–10 055.

[21] J. Wu and M. Harandi, "Munba: Machine unlearning via nash bargaining," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 4754–4765.

[22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[23] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[24] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," 2009.

[25] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein *et al.*, "Imagenet large scale visual recognition challenge," *International journal of computer vision*, vol. 115, no. 3, pp. 211–252, 2015.

[26] Y. Zhong and W. Deng, "Face transformer for recognition," *arXiv preprint arXiv:2103.14803*, 2021.

[27] Y. Luo, X. Zhao, H. Shen, C. Fu, and H. Jiang, "A novel deep learning network for iris recognition using non-global interactive channel attention and lightweight spatial attention," *Signal, Image and Video Processing*, vol. 19, no. 11, p. 950, 2025.

[28] X. Guan, Z. Pan, J. Feng, and J. Zhou, "Joint identity verification and pose alignment for partial fingerprints," *IEEE Transactions on Information Forensics and Security*, 2024.

[29] S. Kadhim, J. K. S. Paw, C. T. Yaw, S. Ameen, and A. Alkhayyat, "Deep learning for robust iris recognition: Introducing synchronized spatiotemporal linear discriminant model-iris." *Adv. Artif. Intell. Mach. Learn.*, vol. 5, no. 1, pp. 3446–3464, 2025.

[30] X. Sun, C. Wang, Y. Wang, J. Wei, and Z. Sun, "Irisformer: A dedicated transformer framework for iris recognition," *IEEE Signal Processing Letters*, 2024.

[31] J. Yu, L. Niu, C. Gao, Z. Cao, and H. Zhao, "Partial fingerprint matching via feature similarity and pre-training," in *2024 IEEE international joint conference on biometrics (IJCB)*. IEEE, 2024, pp. 1–9.

[32] S. Wu, Z. Huang, C. Zhang, C. Wang, H. Chen *et al.*, "Privacy-preserving face recognition with a mask-encoded microlens array," *Advanced Imaging*, vol. 2, no. 1, p. 011001, 2025.

[33] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *International conference on machine learning*. PMLR, 2021, pp. 10 347–10 357.

[34] M. Singh, S. Nagpal, M. Vatsa, R. Singh, A. Noore, and A. Majumdar, "Gender and ethnicity classification of iris images using deep class-encoder," in *2017 IEEE international joint conference on biometrics (IJCB)*. IEEE, 2017, pp. 666–673.

[35] Y. I. Shehu, A. Ruiz-Garcia, V. Palade, and A. James, "Sokoto coventry fingerprint dataset," *arXiv preprint arXiv:1807.10609*, 2018.

[36] D. Yi, Z. Lei, S. Liao, and S. Z. Li, "Learning face representation from scratch," *arXiv preprint arXiv:1411.7923*, 2014.

[37] C. A. Choquette-Choo, F. Tramer, N. Carlini, and N. Papernot, "Label-only membership inference attacks," in *International conference on machine learning*. PMLR, 2021, pp. 1964–1974.

[38] T. Shibata, G. Irie, D. Ikami, and Y. Mitsuzumi, "Learning with selective forgetting." in *IJCAI*, vol. 3, 2021, p. 4.

[39] A. K. Tarun, V. S. Chundawat, M. Mandal, and M. Kankanhalli, "Fast yet effective machine unlearning," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 9, pp. 13 046–13 055, 2023.

[40] S. M. Ahmed, U. Y. Basaran, D. S. Raychaudhuri, A. Dutta, R. Kundu, F. F. Niloy, B. Guler, and A. K. Roy-Chowdhury, "Towards source-free machine unlearning," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 4948–4957.

[41] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the national academy of sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.

[42] S. Adhikari, V. Kumaravelu, and P. Srijith, "An unlearning framework for continual learning," *arXiv preprint arXiv:2509.17530*, 2025.

[43] B. Liu, Q. Liu, and P. Stone, "Continual learning and private unlearning," in *Conference on Lifelong Learning Agents*. PMLR, 2022, pp. 243–254.

[44] R. Chatterjee, V. Chundawat, A. Tarun, A. Mali, and M. Mandal, "A unified framework for continual learning and unlearning," *arXiv preprint arXiv:2408.11374*, 2024.

[45] Z. Huang, X. Cheng, J. Zhang, J. Zheng, H. Wang, Z. He, T. Li, and X. Huang, "A unified gradient-based framework for task-agnostic continual learning-unlearning," *arXiv preprint arXiv:2505.15178*, 2025.

[46] Y. Wang, J. Wei, C. Y. Liu, J. Pang, Q. Liu, A. P. Shah, Y. Bao, Y. Liu, and W. Wei, "Llm unlearning via loss adjustment with only forget data," *arXiv preprint arXiv:2410.11143*, 2024.

[47] R. Gandikota, J. Materzynska, J. Fiotto-Kaufman, and D. Bau, "Erasing concepts from diffusion models," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 2426–2436.